Project Pipeline Workflow Guide

Objective

Build the MCA mini project 'Data Pipeline Monitoring and Recovery Framework' quickly while understanding each component.

Workflow

Phase 1 – Design (ChatGPT)
Finalize architecture, folder structure, database schema and features.

Phase 2 – Generate Modules (Claude)
Generate only one module at a time:
1. Database connection
2. CSV ETL Pipeline
3. API ETL Pipeline
4. Monitoring Module
5. Recovery Engine
6. Streamlit Dashboard

Never generate the entire project in one prompt.

Phase 3 – Integrate & Debug (ChatGPT)
Run every module, fix errors, integrate components, prepare for reviews and viva.

Development Order

1. PostgreSQL Setup
2. Database Schema
3. CSV Pipeline
4. Monitoring Framework
5. API Pipeline
6. Recovery Engine
7. Streamlit Dashboard
8. Testing & Documentation

Claude Prompt Template

You are an experienced Data Engineer. Generate only one module for my project 'Data Pipeline Monitoring and Recovery Framework'. Use modular Python, PostgreSQL, proper exception handling, logging and comments. Do not generate other files. Assume supporting modules already exist.

Timeline

Day 1: PostgreSQL + Database + CSV Pipeline
Day 2: Monitoring Framework
Day 3: API Pipeline
Day 4: Recovery Engine
Day 5: Streamlit Dashboard
Days 6–7: Testing, polishing and documentation.

Best Practices

- Test one module before starting the next.
- Commit to Git after every completed module.
- Keep screenshots for your report.
- Use ChatGPT for explanations and debugging.
- Use Claude for generating individual modules.